PROGRAM EXECUTION & CONSOLE OUTPUT SHEET

E-Commerce Shopping Cart System — Output Log

Complete Terminal Trace for Normal CRUD Operations, Boundary Checks & Stream Recovery

SUBMITTED BY	SOURCE FILE	COMPILER STANDARD	VERIFICATION STATUS
Vaishali Rathod	Project_Modified.cpp	GCC g++ (C++17)	All 6 Test Cases Passed

Build & Run Command: `$ g++ -std=c++17 -Wall -Wextra Project_Modified.cpp -o shopping_cart && ./shopping_cart`

01 INTERACTIVE CONSOLE EXECUTION TRACE

Session Part A: Adding & Displaying Items [stdout / stdin]

```
===== E-Commerce Shopping Cart ===== 1. Add Item 2. Update Quantity 3. Remove Item 4. Display Cart 5. Exit Choice: 4 Cart is empty. ===== E-Commerce Shopping Cart ===== 1. Add Item 2. Update Quantity 3. Remove Item 4. Display Cart 5. Exit Choice: 1 ID: 101 Name: Wireless Mechanical Keyboard Price: 3499.50 Qty: 1 Item added to cart. Choice: 1 ID: 102 Name: Optical Gaming Mouse Price: 1250.00 Qty: 2 Item added to cart. Choice: 1 ID: 103 Name: USB-C Hub Multiport Adapter Price: 1899.00 Qty: 1 Item added to cart. Choice: 4 --- Cart Details --- ID: 101 | Name: Wireless Mechanical Keyboard | Price: Rs 3499.50 | Qty: 1 ID: 102 | Name: Optical Gaming Mouse | Price: Rs 1250.00 | Qty: 2 ID: 103 | Name: USB-C Hub Multiport Adapter | Price: Rs 1899.00 | Qty: 1 Total Cost: Rs 7898.50
```

Session Part B: Validation, Update & Removal [stdout / stdin]

```
Choice: 1 ID: 101 Name: Duplicate Item Check Price: 999.00 Qty: 1 A product with this ID already exists. Choice: 1 ID: 104 Name: Laptop Stand Price: -450.00 Qty: 0 Invalid product details. ID must be positive, price cannot be negative, and quantity must be greater than zero. Choice: 2 ID: 102 New Qty: 3 Quantity updated. Choice: 3 ID: 101 Item removed. Choice: xyz Invalid input. Please enter a number from 1 to 5. Choice: 4 --- Cart Details --- ID: 102 | Name: Optical Gaming Mouse | Price: Rs 1250.00 | Qty: 3 ID: 103 | Name: USB-C Hub Multiport Adapter | Price: Rs 1899.00 | Qty: 1 Total Cost: Rs 5649.00 Choice: 5 Exiting...
```

02 OUTPUT VERIFICATION & TEST CASE MATRIX

Test ID	Operation Tested	Input Parameters	Console Output Verified	Result
OUT-01	Empty Cart Check	`Choice: 4` (Initial state)	Prints `Cart is empty.`	PASSED
OUT-02	Tail Node Insertion	IDs 101, 102, 103	Prints `Item added to cart.`; billing = `Rs 7898.50`	PASSED
OUT-03	Duplicate ID Prevention	Re-insert existing `ID: 101`	Prints `A product with this ID already exists.`	PASSED
OUT-04	Domain Constraint Guard	`Price: -450.00, Qty: 0`	Prints `Invalid product details...`	PASSED
OUT-05	Quantity Update & Head Delete	Update 102→3; Remove 101	Recalculates total cleanly to `Rs 5649.00`	PASSED